Revision Week

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[IN PREPARATION - CHECK BACK LATER]

Introduction

This session

We are now halfway through the module! Our timetabled session this week represents a chance to recap tricky concepts at your request, and then practice applying your new knowledge.

This worksheet has been designed in a way that reflects how you will need to use your knowledge in the exam, but with a more practical flavour. You might not get through all the exercises today (especially if we spend some time recapping content together), but you can use it as a revision tool and ask questions on the VLE.

The mock exam will be released on the VLE after the timetabled session (6pm, Monday), and will be available for you to do in your own time.

Approaching the worksheet

The worksheet will walk you through different analyses applied to a central (fictional) dataset. Where relevant, you can take your pick from three levels of R enthusiasm:

1. **(Wannabee) R Whizz!** Use your notes and analysis scripts from previous weeks to perform all the tasks from scratch, without looking at the code suggestions. Recommended for people who are independently motivated to learn and develop R skills.
2. **R Curious:** Use the code suggestions to set you on your way for each task. Recommended for people who are enjoying learning R but don't want to get held up at the cost of making it through the worksheet.
3. **R Overwhelmed:** Copy and paste the provided code to generate output where necessary. Recommended for people who want to save learning R for another time.

Fictional context

Today's dataset is simulated for the purpose of demonstrating the different analytic choices you might experience for any one project. Our central question is: ***What helps students cope with stress, sleep better, and perform well academically?***

Imagine that you are part of a university research team investigating student wellbeing. From your reading, you find that previous studies seem to disagree about which interventions work. You suspect that individual students might respond differently to interventions, and that the average effect may conceal important differences.

Set-up

- Open you ARM_Practicals.RProj file to get oriented in RStudio.
- Download the data from the VLE, and save it in your data folder as usual.
- Create a new R script for today's session, and save it in your scripts folder with a sensible name (e.g., ARM6_Revision)
- At the top of your script, load the following packages: *tidyverse, metafor, lme4, lmerTest, quantreg.* Note that you may need to install them first if you are working on a new device.

```
# If you need to install any packages... e.g.,
# install.packages("package")

# Load packages
library(tidyverse)
library(metafor)
library(lme4)
library(lmerTest)
library(quantreg)
```

Part 1

A sensible first step in the

Meta-analysis: Do mindfulness interventions reduce student stress across published studies?

1. Which type of meta-analysis would be appropriate for these data, and why?
2. Take a look at the forest plot:
    1. How many of the original studies showed a significant benefit of the intervention?
    2. Which studies were given most and least weight in the model?
    3. Based on the existing literature, do interventions seems to work?
3.

Part 2

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- Mixed-effects models: How does stress change over time during an intervention, accounting for repeated measurements and differences between students?
- Quantile regression: Does an intervention help students with exceptionally high stress as much as those with more typical stress levels?
- Bootstrapping: How stable are the estimated intervention effects or indirect effects when their sampling distributions are uncertain?

Why it works: students understand the problem immediately, and the context naturally motivates the need for more than one statistical approach.